

WTW158 Calculus

8

Class Test 2 (Mo)

24-28 February 2025

Time: 20 min

Surname:

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Initials:

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Student number:

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Group:

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M-number

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Use a pen to complete the test. Sketches must also be in pen. Simplify all your answers. Use interval notation, where applicable. No calculators may be used.

QUESTION 1 [2 marks] Indicate whether the statement is true (T) or false (F).

(a) The function $f(x) = |\sin x|$ is odd.

F

(b) When $f(x) + f(-x) = 0$, we say f is an even function.

F

(c) The function $f(x) = \sin x$ is increasing on the interval $(\frac{3\pi}{2}, \frac{5\pi}{2})$.

T

(d) The function $f(x) = \frac{\pi^x}{x^2+1}$ is a rational function.

F

QUESTION 2 [2 marks]

Solve for x if $2\cos^2 x + \sin x = 1$.

$$2(1 - \sin^2 x) + \sin x = 1$$

$$\Rightarrow 2 - 2\sin^2 x + \sin x = 1$$

$$\Rightarrow 1 - 2\sin^2 x + \sin x = 0$$

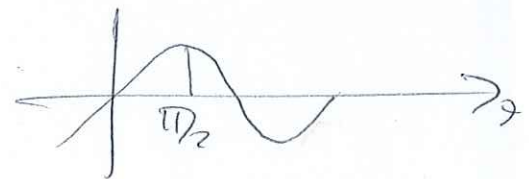
$$\Rightarrow 1 + \sin x - 2\sin^2 x = 0$$

$$\Rightarrow (1 + 2\sin x)(1 - \sin x) = 0$$

$$\Rightarrow \sin x = -\frac{1}{2} \quad \text{or} \quad \sin x = 1$$

$$x = 2\pi - \frac{\pi}{6} + 2k\pi$$

$$\text{or } x = \pi + \frac{\pi}{6} + 2k\pi, k \in \mathbb{Z}$$

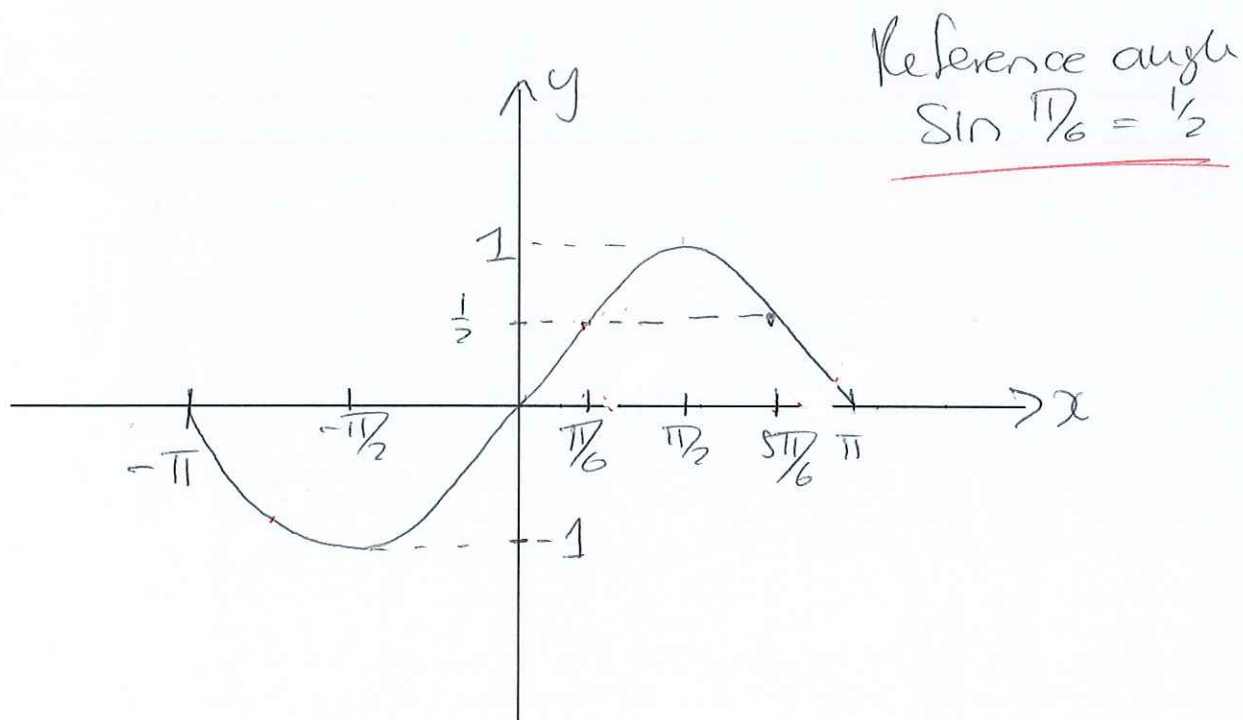


$$\Rightarrow x = \frac{11\pi}{6} + 2k\pi$$

$$\text{or } x = -\frac{\pi}{6} + 2k\pi \quad \text{or} \quad x = \frac{\pi}{2} + 2k\pi, k \in \mathbb{Z}$$

QUESTION 3 [4 marks]

Use a graph to solve $\sin x < \frac{1}{2}$, $x \in [-\pi, \pi]$. Indicate all intercepts and relevant information on the graph and write your answer in interval notation.



$$\Rightarrow x \in [-\pi, \frac{\pi}{6}) \cup (\frac{5\pi}{6}, \pi]$$

OK

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Class Test 2 (Tu)

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Use a pen to complete the test. Sketches must also be in pen. Simplify all your answers. Use interval notation, where applicable. No calculators may be used.

QUESTION 1 [2 marks] Indicate whether the statement is true (T) or false (F).

(a) The function $f(x) = x|x|$ is even. F

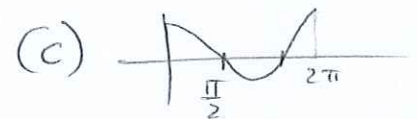
(b) When $f(x) - f(-x) = 0$, we say f is an even function. T

(c) The function $f(x) = \cos x$ is increasing on the interval $(\frac{3\pi}{2}, \frac{5\pi}{2})$. F

(d) The function $f(x) = \frac{\pi^x}{x^2+1}$ is a rational function. F

$$\begin{aligned} f(-x) &= -x|-x| \\ &= -x|x| \\ &= -f(x) \end{aligned}$$

$$(b) f(x) = f(-x)$$



QUESTION 2 [2 marks]

Solve for $x \in [0, 2\pi)$ if $\cos(2x) \cos^2 x = 2 \cos(2x)$.

$$\cos 2x \cos^2 x - 2 \cos 2x = 0$$

$$\Rightarrow \cos 2x (\cos^2 x - 2) = 0$$

$$\text{Since } \cos^2 x \neq 2, \cos 2x = 0$$

$$\Rightarrow 2x = \frac{\pi}{2} + 2k\pi, k \in \mathbb{Z}$$

$$\text{or } 2x = \frac{3\pi}{2} + 2k\pi, k \in \mathbb{Z}$$

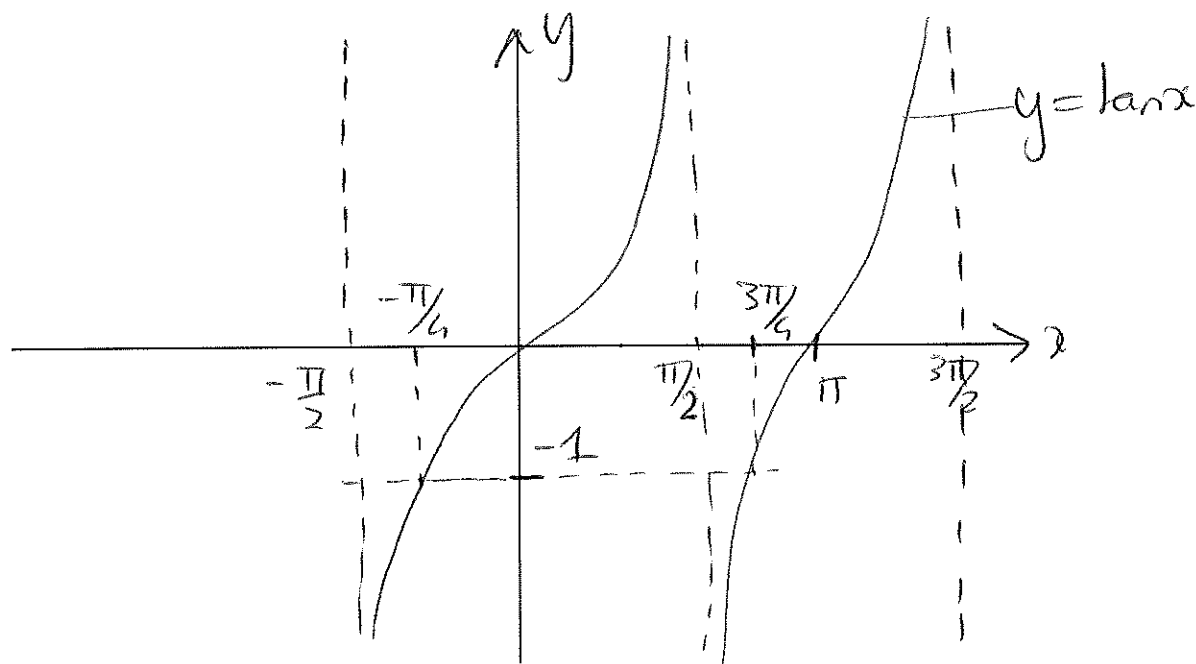
$$\Rightarrow \left(x = \frac{\pi}{4} + k\pi, k \in \mathbb{Z} \right) \text{ or } \left(x = \frac{3\pi}{4} + k\pi, k \in \mathbb{Z} \right), \text{ But } x \in [0, 2\pi)$$

$$\Rightarrow x \in \left\{ \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4} \right\}$$

QUESTION 3 [4 marks]

Use a graph to solve $\tan x < -1$, $x \in (-\frac{\pi}{2}, \frac{3\pi}{2})$. Indicate all intercepts and relevant information on the graph and write your answer in interval notation.

$$\tan \frac{\pi}{4} = 1 \Rightarrow \text{reference angle: } \frac{\pi}{4}.$$



$$x \in (-\frac{\pi}{2}, -\frac{\pi}{4}) \cup (\frac{\pi}{2}, \frac{3\pi}{4})$$

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Use a pen to complete the test. Sketches must also be in pen. Simplify all your answers. Use interval notation, where applicable. No calculators may be used.

QUESTION 1 [2 marks] Indicate whether the statement is true (T) or false (F).

(a) When $f(x) - f(-x) = 0$, we say f is an even function. T

(b) The function $f(x) = \sin x$ is increasing on the interval $(\frac{3\pi}{2}, \frac{5\pi}{2})$. T

(c) The power function $f(x) = x^n, n \in \{1, 3, 5, \dots\}$ is an odd function. T

(d) The function $f(x) = \frac{a^x}{x^2+1}, a > 0$ is a rational function. F

QUESTION 2 [2 marks]

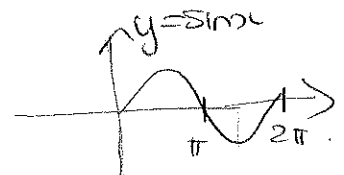
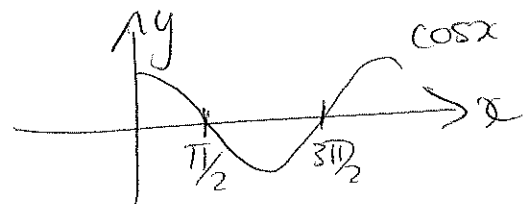
Solve for $x \in [0, 2\pi)$ if $2 \cos x + \sin(2x) = 0$.

$$2\cos x + 2\sin x \cos x = 0$$

$$\Rightarrow 2\cos x(1 + \sin x) = 0$$

$$\Rightarrow \cos x = 0 \text{ or } \sin x = -1$$

$$x = \frac{\pi}{2} \text{ or } \frac{3\pi}{2} \text{ or } x = \frac{3\pi}{2}$$



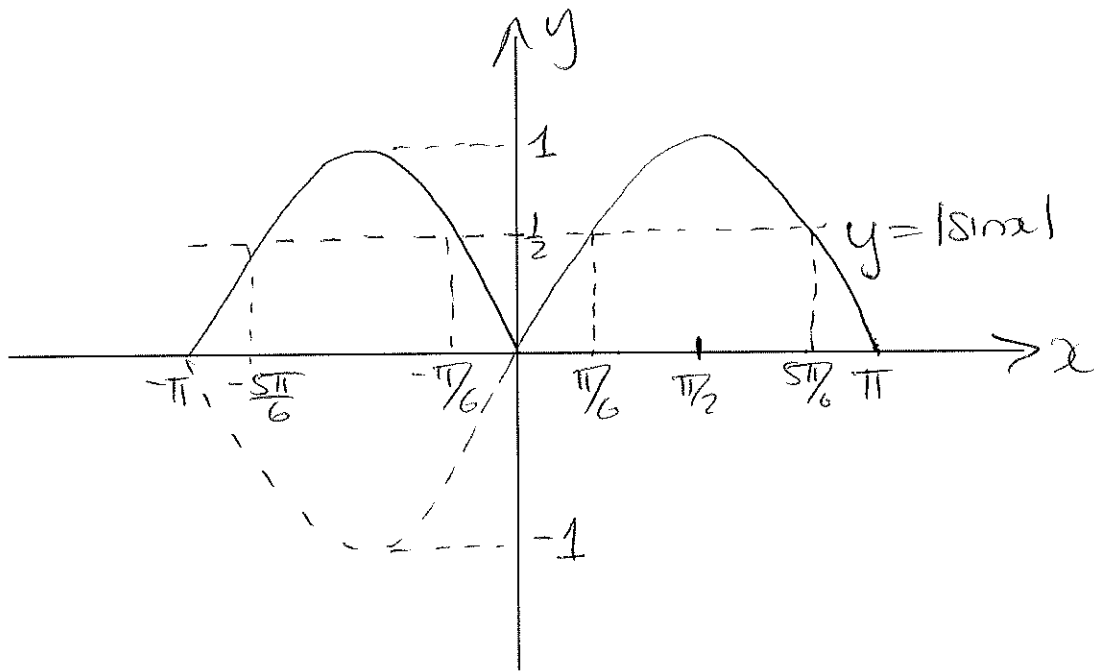
$$\Rightarrow x \in \left\{ \frac{\pi}{2}, \frac{3\pi}{2} \right\}$$

$$\text{OR: } x = \frac{\pi}{2}, \frac{3\pi}{2}$$

QUESTION 3 [4 marks]

Use a graph to solve $|\sin x| \geq \frac{1}{2}$, $x \in [-\pi, \pi]$. Indicate all intercepts and relevant information on the graph and write your answer in interval notation.

Reference angle: $\sin x = \frac{1}{2}$
 $\Rightarrow x = \frac{\pi}{6}$



$$-\pi + \frac{\pi}{6} = -\frac{5\pi}{6}$$

$$\pi - \frac{\pi}{6} = \frac{5\pi}{6}$$

$$\Rightarrow x \in \left[-\frac{5\pi}{6}, -\frac{\pi}{6}\right] \cup \left[\frac{\pi}{6}, \frac{5\pi}{6}\right].$$

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QUESTION 1 [2 marks] Indicate whether the statement is true (T) or false (F).

- (a) The function $f(x) = \frac{x}{x+1}$ is odd. *False*
- (b) When $f(x) - f(-x) = 0$, we say f is an even function. *True*
- (c) The function $f(x) = \sin x$ is increasing on the interval $(\frac{3\pi}{2}, \frac{5\pi}{2})$. *True*
- (d) The function $f(x) = x$ is both a power function and a polynomial. *True*

QUESTION 2 [2 marks]

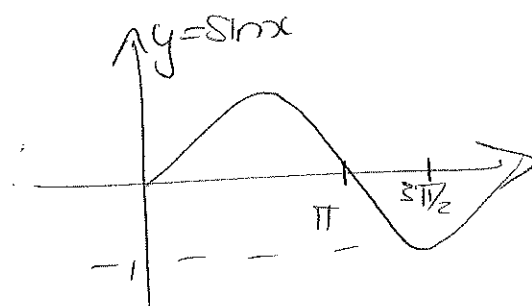
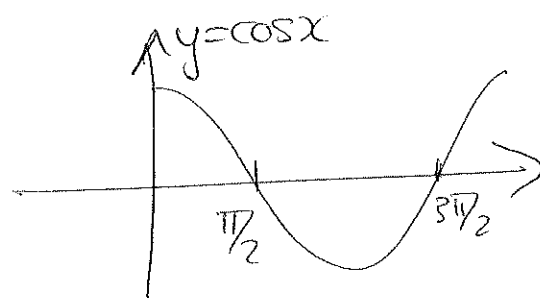
Solve for $x \in [0, 2\pi)$ if $2 \cos x + \sin(2x) = 0$.

$$2 \cos x + 2 \sin x \cos x = 0$$

$$\Rightarrow 2 \cos x (1 + \sin x) = 0$$

$$\Rightarrow \cos x = 0 \text{ or } \sin x = -1$$

$$x = \frac{\pi}{2} \text{ or } x = \frac{3\pi}{2}$$



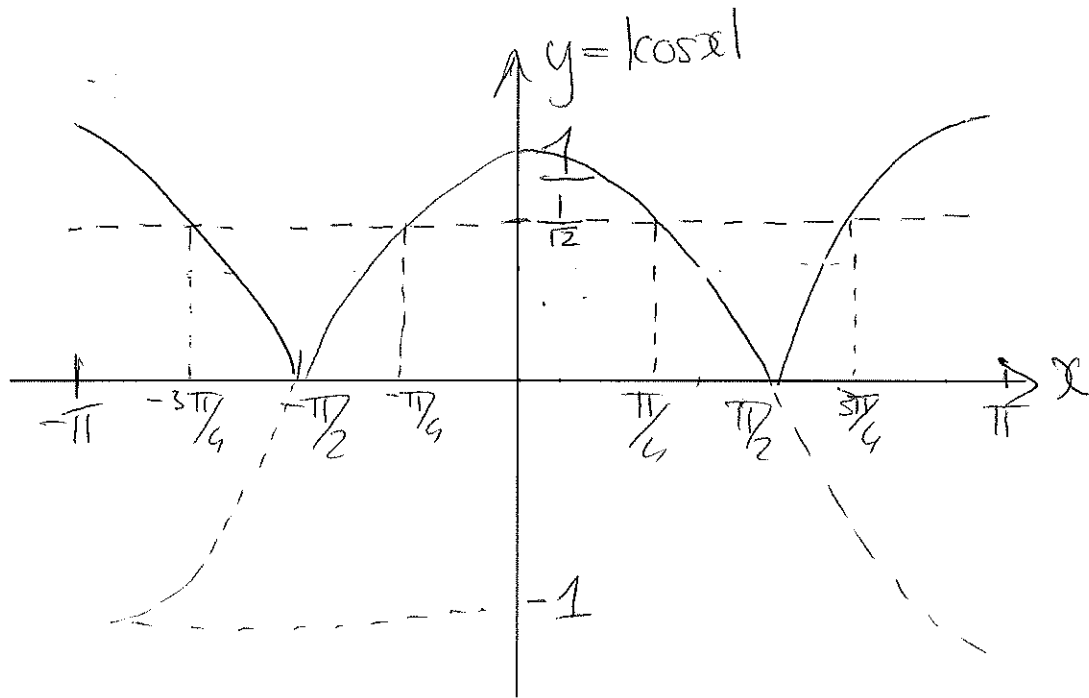
$$\Rightarrow x \in \left\{ \frac{\pi}{2}, \frac{3\pi}{2} \right\}$$

Ans: $x = \frac{\pi}{2} \text{ or } x = \frac{3\pi}{2}$

QUESTION 3 [4 marks]

Use a graph to solve $|\cos x| > \frac{1}{\sqrt{2}}$, $x \in [-\pi, \pi]$. Indicate all intercepts and relevant information on the graph and write your answer in interval notation.

Reference angle
 $\cos \frac{\pi}{4} = \frac{1}{\sqrt{2}}$
 $\Rightarrow$ Ref. angle: $\frac{\pi}{4}$.



$$x \in [-\pi, -3\pi/4) \cup (-\pi/4, \pi/4) \cup (3\pi/4, \pi]$$